

*SPW2plus*

Revision 1.0

# User's Manual

Single socket PROM/OTP/MTP programmer

August 1, 2000



# 1. General description and Features

## 1.1 Main features

*SPW2plus is a high-speed single socket programmer for the Samsung serial / Parallel OTP and MTP(Flash memory MCU) devices. It can program one device at a time in Samsung standard DIP, SOP, QFP or TQFP on PC-based mode. Main features are as follows:*

- PC-based Menu-driven software support for simple operation
- Very fast program and verify time: about 0.8 second per 1K byte
- Load file, Buffer dump, Read device, Program, Verify, Read protection, Device Checksum, Buffer Checksum, and Blank check functions for OTP and MTP devices. In addition, Erase, HardLock, and LDC Protection functions are also available for MTP devices.
- Auto data increment & writing for serial or secret numbers of product
- Window 95/98/2000/NT operating system
- Maximum buffer memory is 1M bytes(1024K bytes)
- Convenient serial port connection to any IBM compatible PC or Notebook.
- AC110/220V to 15DC power adapter included
- Full line-ups programming socket adapter for Samsung OTPs and MTPs.  
( Sold separately, See C & A Technology Home page: [www.cnatech.com](http://www.cnatech.com) )

## 1.2 Packing Includes

- (1) SPW2plus Installation diskette
- (2) SPW2plus Main body
- (3) RS-232C Cable
- (4) Power Adapter(110/220VAC to 15VDC )
- (5) User's Manual

## 1.3 Host system Requirements

- (1) IBM or compatible 486/Pentium with minimum 8MB extended memory.
- (2) Windows95/98/2000/NT operating system with 5MB of free hard disk space.
- (3) One RS-232C serial port (COM1~COM4, 9 Pin type).
- (4) 3.5" floppy disk drive.

(Note) Programming adapter socket is needed depending on package type of OTP/MTP devices and sold separately. (See Appendix A, B, C, D adapter socket table)

## 1.4 Device support

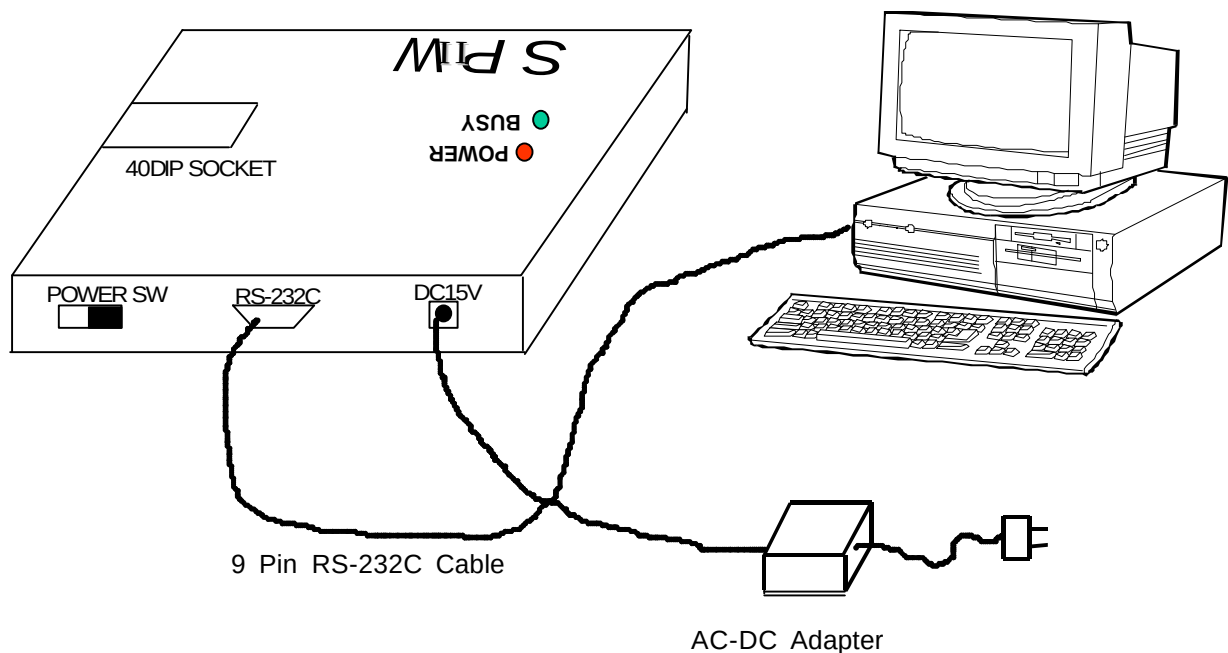
- (1) SPW2plus supports all of Samsung Serial/Parallel type OTP and MTP (Flash memory based MCU) devices.
- (2) Standard EPROM: 27(C)256, 27(C)512

## 1.5 Software installation

- (1) Insert the installation disk #1 into floppy disk drive.
- (2) Execute "Setup.exe" file on the installation disk #1.
- (3) Follow the instruction message on the screen. When the installation is finished, the program icon of the SPW2plus will be generated in the start/programs menu.

## 1.6 Hardware setup

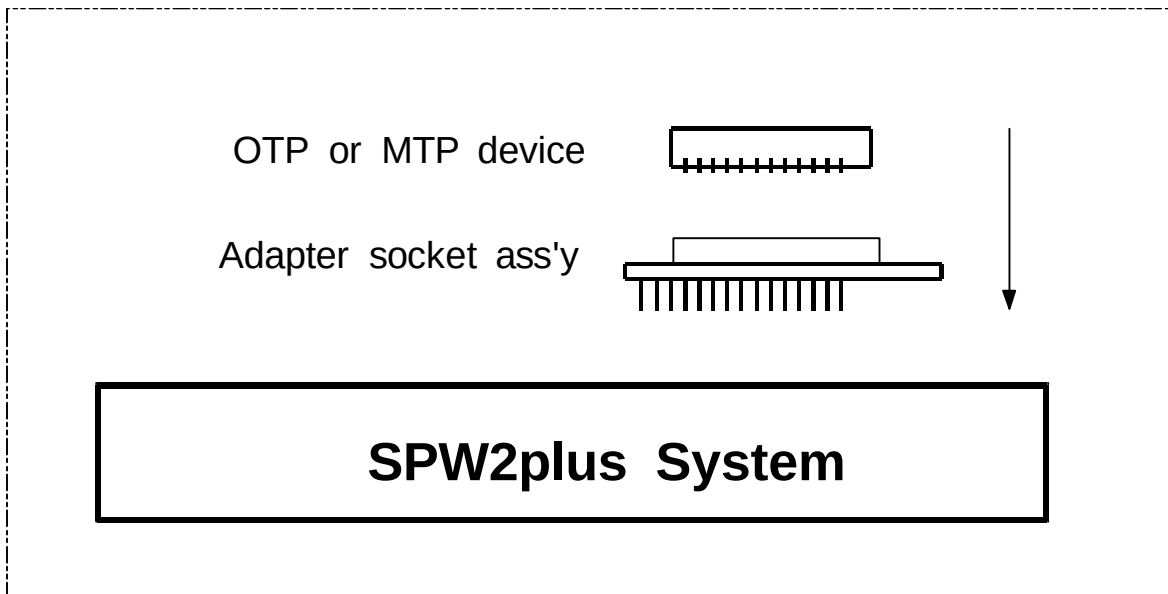
- (1) Set as figure below.
- (2) Set the voltage selection switch(110/220V switch) of the power adapter depending on your country's power system.
- (3) Turn on the SPW2plus system power, and check if the BUSY LED blinks three times. If the LED does not blink, turn power off and turn on again a few minutes later.



## 1.7 How to mount devices on the SPW2plus system

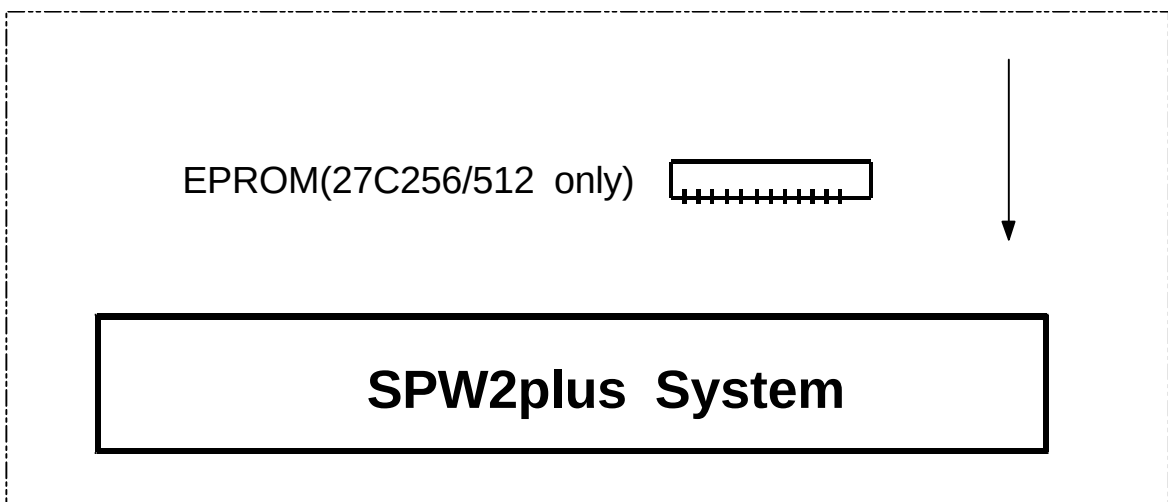
### (1) OTP/MTP

*Every OTP/MTP needs an adapter socket for OTP/MTP programming, you choose an adapter socket and setup as follow. (See Appendix A, B, C, D adapter socket table)*



### (2) EPROM

*For EPROM programming, the adapter socket is not needed.*

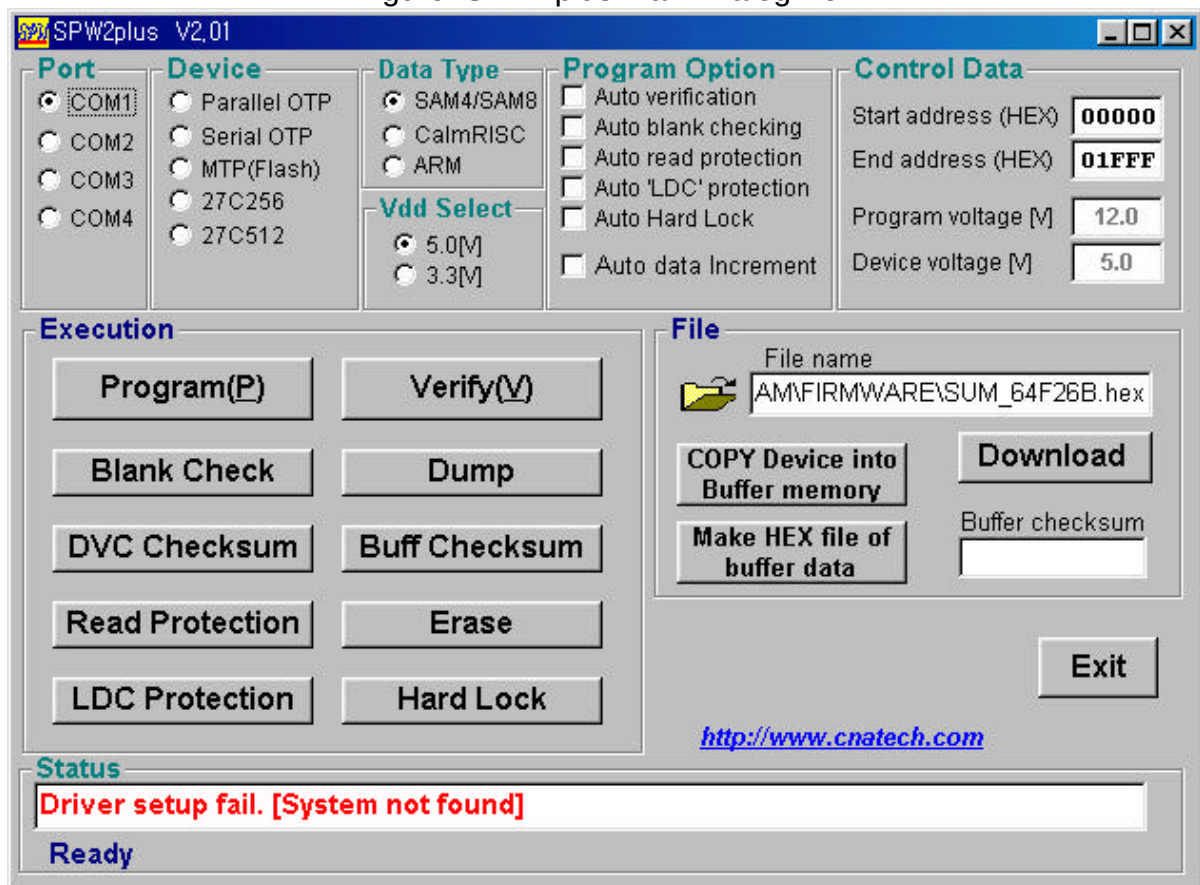


## 2. Operation descriptions

### 2.1 Running SPW2plus software

- (1) Start up SPW2plus
- (2) Turn on the power switch of SPW2plus System.
- (3) Click SPW2plus icon under the programs on Win95/98/2000/NT start menu.

Figure. SPW2plus Main Dialog Box



After running SPW2plus, the above window will be appeared.  
If the port or device was not selected, you should select the corresponding port first and device type.

## 2.2 Operation Menu

### (1) Device

- .. **Parallel OTP:** Select this item if you want to use the parallel type OTP such as KS88P0916(S3P8095), KS88P0424(S3P8047), and KS88P01424(S3P80E7).
- .. **Serial OTP:** Select this item if you want to use the serial type OTP.
- .. **MTP(Flash):** Select this item if you want to use the serial type MTP(Flash type program memory based MCU).
- .. **27C256/512:** Select this item if you want to use EPROM.

### (2) Data Type

*This selection should be decided in accordance with the ROM data type of devices. The data type selection is available only when MTP is selected in the device selection menu.*

- .. **SAM4/SAM8:** Select this item if device is 8-bit word of ROM format such as KS51P860(S3P1860), SAM4(KS57 series, or S3P7xxx series), and SAM8 (KS86/88 series, S3P8/9xxx, or S3F8/9xxx series).
- .. **CalmRISC:** Select this item if device is 16-bit word of ROM format such as CalmRISC (KS85 series, S3PBxxx, or S3FBxxx series).
- .. **ARM:** Select this item if device is 32-bit word of ROM format such as ARM7 (KS17/32 series, S3P3/4xxx, S3F3/4xxx series).

### (3) Vdd Select

*This selection should be decided in accordance with operating voltage of devices. This selection is available only when MTP is selected in the device selection menu.*

- .. **5.0[V]:** Select this item if devices are the 5-volt of operating voltage.
- .. **3.3[V]:** Select this item if devices are the 3.3-volt of operating voltage.

### (4) Program Option

*This Check box helps to program OTP/MTP simply.*

- .. **Auto verification:** After programming, verify the device automatically.
- .. **Auto blank checking:** Before programming, check the device is blank or not.
- .. **Auto read protection:** After programming, makes it unreadable.
- .. **Auto 'LDC' protection:** After programming, makes it as LDC instruction protection. This option is available when MTP device is selected in Device menu.
- .. **Auto HARD LOCK:** After programming, make it as hard lock. This option is available when MTP device is selected in Device menu. (HARDLOCK is used to protect programmed device from rewriting by electrical damage.)
- .. **Auto data Increment:** Every programming, automatically increase the writing data in number one specified by the base address and offset value. This option is a special purpose when you want to write the different serial or secret numbers into product automatically every programming. Please be careful to use this

option because it may change data of buffer that had been downloaded or read from the master device before.

#### [ Writing Data text box ]

Please enter the initial data to be programmed into the first device of OTP/MTPs. This data value will be increased by one automatically after programming. The digit of this writing data is also limited by value of Range of Address (Offset). For example:

- Range of Address(Offset):0           -> Writing data: 00~FF
- Range of Address(Offset):1(or -1) -> Writing data: 0000~FFFF
- Range of Address(Offset):3(or -3) -> Writing data: 00000000~FFFFFFFF

#### [ Base address text box ]

Please enter the base address of OTP/MTP device that least significant byte (LSB) of the writing data is programmed. The next byte of the writing data will be programmed at base address plus one(Base address+1). For example:

< Assumption > Range of Address(Offset):+1

Writing data: 0001(Initial data)

Base Address: 00F000

< 1<sup>st</sup> Program > Address 00F000: data 01h,

Address 00F001: data 00h, are programmed and the writing data will be increased as 0002h automatically.

< 2nd Program > Address 00F000: data 02h,

Address 00F001: data 00h, the writing data will be 0003h.

...

< 65535th Program > Address 00F000: data FFh,

Address 00F001: data FFh, the writing data will be 0000h.

#### [ Range of address(Offset) text box ]

Please enter the range of address from -7 to +7 offset value that decides maximum address of area to be programmed with the writing data. The "-" sign means direction to the decreasing address from the base address. For example:

< Assumption > Range of Address (Offset):-2

Writing data: 000201h(Initial data)

Base Address: 00F002h

< 1st Program > Address 00F002: data 01h,

Address 00F001: data 02h,

Address 00F000: data 00h, are programmed and the writing data will be increased as 000202h automatically.

< 2nd Program > Address 00F002: data 02h,

Address 00F001: data 02h.

Address 00F000: data 00h are programmed and the writing data will be increased as 000203h automatically.

...

## (5) Control data

- .. **Start/End address:** Set start/end address for reading/writing range. The end address will be changed to suit the last data address when a data file is downloaded.
- .. **Vpp / Vdd Voltage:** View Vpp/Vdd voltage to check voltage level.

## (6) File

- .. **File Name:** Enter the name of a file to be downloaded. (It can be possible to browse the files using the file open icon.)
- .. **Buffer checksum:** Display the checksum of the buffer memory. Its value will be changed in case of changing the end address of buffer after downloading a file or reading data to the buffer memory.
- .. **Download:** Download a selected file to the buffer memory of SPW2plus system.
- .. **COPY Device into Buffer memory:**  
Copy OTP/MTP/EPROM data to the buffer memory ranged the start and end address.
- .. **Make HEX file of buffer data:**  
Creates a new Intel HEX file by reading the buffer data ranged the start and end address.

## < Hex file format >

There are three kinds of files for downloading to the buffer memory, they are Samsung hex file, Samsung obj file, and Intel standard hex file. Samsung hex file format is generated for debugging by SAMA or SASM assembler. Samsung \*.obj (Object) file is same as Intel hex files for masking order. CalmRISC or ARM code consists of Intel standard hex format. If their file format is binary, you should convert binary into hex format file by using a file conversion utility.

< File example: Samsung Hex file >

```
:05006020100EA0C5D523
:050060229090103030E9
:16000020002600640064006400648320DD0AC482D9830BC5B4
:16000022000000000000000000000000000000000000000010001000101010009030B8
```

Check this point !

< File example: Samsung \*.OBJ file >

```
:10000000000028140CD8081F0CDE881FFCDDCA0A051
:100010008CF0DD13D930DB001B903881F2DCF25C10
:10002000A08100DD0CDD13E6DCAAC8DD068120DC42
:10003000F2DD0EC482DB003381FFCDE88120DCF2EB
```

Check this point !

.. **Intel Standard hex file:**

< File example: Intel standard hex file >

```
:10000000000028140CD8081F0CDE881FFCDDCA0A051
:1000100008CF0DD13D930DB001B903881F2DCF25C10
:100020000A08100DD0CDD13E6DCAAC8DD068120DC42
:100030000F2DD0EC482DB003381FFCDE88120DCF2EB
```



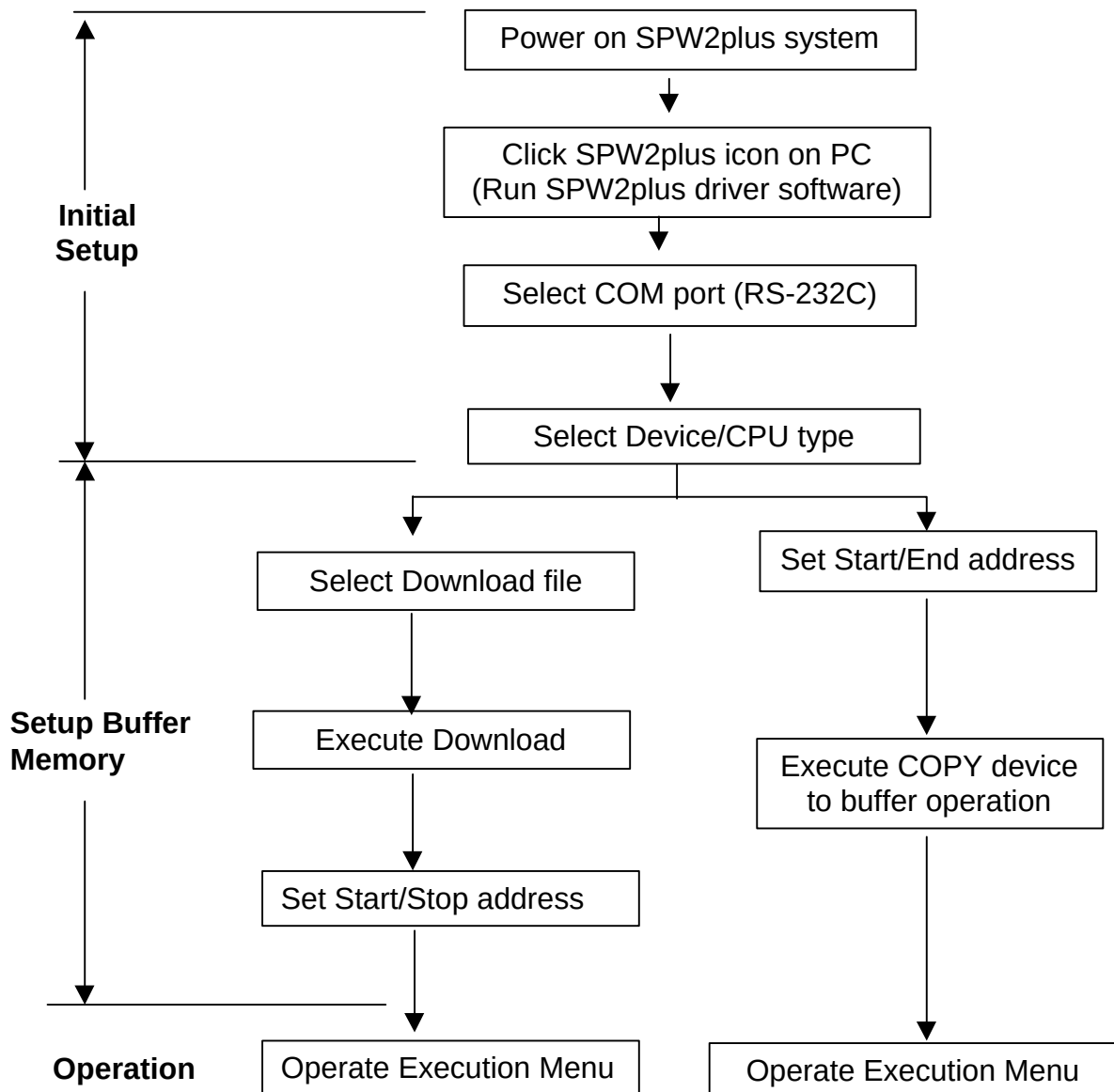
Check this point !

## (7) Execution

- .. **Program:** Program the buffer memory to device (OTP/MTP/EPROM).
- .. **Verify:** Compare the device (OTP/MTP/EPROM) data with the buffer memory.
- .. **Blank Check:** Check if device (OTP/MTP/EPROM) is blank or not.
- .. **Dump:** View the buffer memory or device (OTP/MTP/EPROM) data.
- .. **DVC Checksum:** Get checksum from device (OTP/MTP/EPROM).
- .. **Buffer Checksum:** Get checksum from the buffer memory.
- .. **Read Protection:** Make the OTP/MTP unreadable.
- .. **Erase:** Erase MTP into blank data 'FF'. This command is available for only MTP device.
- .. **LDC Protection:** Make the LDC instruction of MTP not executable. This command is available for only MTP device.
- .. **HARD LOCK:** Make the MTP data not programmable to avoid data change by operator's mistakes or electrical malfunction. This command is available for only MTP device.

\* All execution menus are operated under Start/End address (Except Read Protection, Erase, LDC Protection, and HARDLOCK command).

## 2.3 Operating sequence



## 2.4 TEXTOOL pin configuration.

| Pin No | 27C 512 | 27C 256 | Parallel OTP | Serial OTP/ MTP | Serial OTP/ MTP | Parallel OTP | 27C 256 | 27C 512 | Pin No |
|--------|---------|---------|--------------|-----------------|-----------------|--------------|---------|---------|--------|
| 1      |         |         |              | SCLK            | RESET           |              |         |         | 40     |
| 2      |         |         |              | SDAT            |                 |              |         |         | 39     |
| 3      |         |         |              | VPP             |                 |              |         |         | 38     |
| 4      |         |         |              | VDD             |                 | MEMREGB      |         |         | 37     |
| 5      |         |         | A15          |                 |                 |              |         |         | 36     |
| 6      |         |         | A14          |                 |                 | OTPCE-       |         |         | 35     |
| 7      | A15     | VPP     | VPP          |                 |                 | VDD          | VDD     | VDD     | 34     |
| 8      | A12     | A12     | A12          |                 |                 | PGMB         | A14     | A14     | 33     |
| 9      | A7      | A7      | A7           |                 |                 | A13          | A13     | A13     | 32     |
| 10     | A6      | A6      | A6           |                 |                 | A8           | A8      | A8      | 31     |
| 11     | A5      | A5      | A5           |                 |                 | A9           | A9      | A9      | 30     |
| 12     | A4      | A4      | A4           |                 |                 | A11          | A11     | A11     | 29     |
| 13     | A3      | A3      | A3           |                 |                 | OEB          | OEB     | VPP     | 28     |
| 14     | A2      | A2      | A2           |                 |                 | A10          | A10     | A10     | 27     |
| 15     | A1      | A1      | A1           |                 |                 |              | CE      | CE      | 26     |
| 16     | A0      | A0      | A0           |                 |                 | D7           | D7      | D7      | 25     |
| 15     | D0      | D0      | D0           |                 |                 | D6           | D6      | D6      | 24     |
| 18     | D1      | D1      | D1           |                 |                 | D5           | D5      | D5      | 23     |
| 19     | D2      | D2      | D2           |                 |                 | D4           | D4      | D4      | 22     |
| 20     | GND     | GND     | GND          | GND             |                 | D3           | D3      | D3      | 21     |

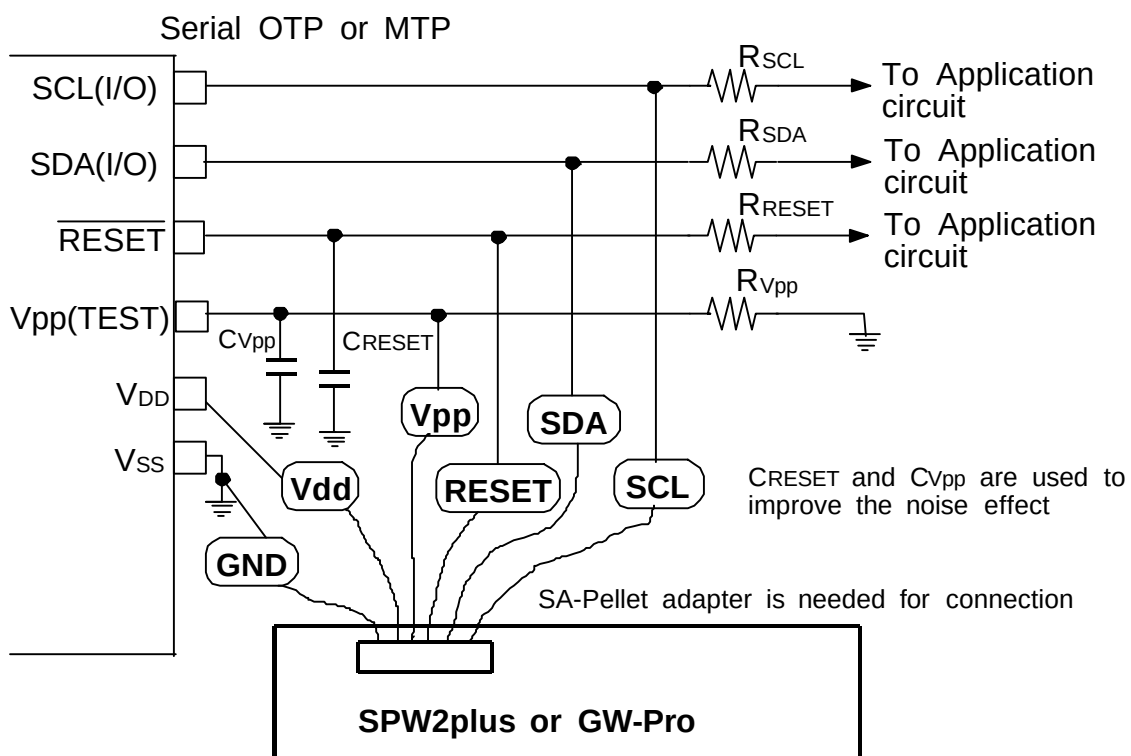
### 3. On-board writing application

Many users often use the pellet devices, but the pellet devices can not be programmed alone, in that case the users need to use the ON-BOARD writing method. SAMSUNG OTP/MTP (serial type) needs only 6 signal lines including VDD and GND pins. Therefore the on-board writing is possible if the writing signal lines are considered when the PCB of application board is designed.

#### < Circuit design guide >

At the OTP/MTP writing, the SPW2plus (or Smart Kit) needs 6 signal lines that are GND, VDD, RESET, TEST, SDA and SCL. When you design the PCB circuits, you should consider the usage of these signal lines for the on-board writing.

In case of TEST pin, normally test pin is connected to GND but in writing mode the programming voltage ( $V_{pp}=12V$ ) power signal is input to this pin. Therefore, combining these two cases, a resistor should be inserted between the TEST pin and GND. The Reset, SDA and SCL should be treated under the same consideration. Please be careful to design the related circuit of these signal pins because rising/falling timing of  $V_{pp}$ , SCL and SDA is very important for proper programming.



## &lt; Reference Table for Connection &gt;

| Pin name  | I/O mode<br>in Applications | Resistor<br>(need) | Required Value                                                   |
|-----------|-----------------------------|--------------------|------------------------------------------------------------------|
| Vpp(TEST) | Input                       | Yes                | $R_{VPP}$ is 10 Kohm ~ 50 Kohm<br>$C_{VPP}$ is 0.01uF ~ 0.02uF   |
| RESET     | Input                       | Yes                | $R_{RESET}$ is 2 Kohm ~ 5 Kohm<br>$C_{RESET}$ is 0.01uF ~ 0.02uF |
| SDA(I/O)  | Input                       | Yes                | $R_{SDA}$ is 2 Kohm ~ 5 Kohm                                     |
|           | Output                      | No(Note)           |                                                                  |
| SCL(I/O)  | Input                       | Yes                | $R_{SCL}$ is 2 Kohm ~ 5 Kohm                                     |
|           | Output                      | No(Note)           |                                                                  |

(Note) In on-board writing mode, very high-speed signal will be provided to pin SCL and SDA. And it will be cause some damages to the application circuits connected to SCL or SDA port if the application circuit is designed as high speed response such as relay control circuit.

## &lt; Appendix A &gt;

SAM4(KS57Pxxxx, S3P7xxx Series) adapter socket table

| <i>Device name</i>  |                          | <i>Package type</i> | <i>Adapter socket (New name)</i>                                 | <i>Remark</i> |
|---------------------|--------------------------|---------------------|------------------------------------------------------------------|---------------|
| <i>New</i>          | <i>Old</i>               |                     |                                                                  |               |
| S3P7048             | KS57P0408                | 42SDIP<br>44QFP     | SA57P0408-42SDIP(SA7048-42SDIP)<br>SA57P0408-44QFP(SA7048-44QFP) |               |
| S3P7054<br>S3P70F4  | KS57P0504<br>KS57P01504  | 30SDIP<br>32SOP     | SA-30SDIP<br>SA-32SOP                                            | Standard type |
| S3P7065             | KS57P0616                | 42SDIP<br>44QFP     | SA-42SDIP<br>SA-44QFP                                            | Standard type |
| S3P72B9<br>S3P72M9  | KS57P21132<br>KS57P22132 | 128QFP              | SA-128QFP                                                        | Standard type |
| S3P72C8             | KS57P21208               | 42SDIP<br>44QFP     | SA-42SDIP<br>SA-44QFP                                            | Standard type |
| S3P72G9             | KS57P21632               | 100QFP              | SA-100QFP                                                        | Standard type |
| S3P72H8             | KS57P21708               | 64QFP               | SA-64QFP                                                         | Standard type |
| S3P72I9             | KS57P21832               | 100QFP              | SA-100QFP                                                        | Standard type |
| S3P7234<br>S3P72N4  | KS57P2304<br>KS57P22304  | 64QFP               | SA-64QFP                                                         | Standard type |
| S3P7235<br>S3P72N5  | KS57P2316<br>KS57P22316  | 80QFP               | SA-80QFP                                                         | Standard type |
| S3P7295             | KS57P2916                | 80QFP               | SA-80QFP                                                         | Standard type |
| S3P72E8<br>S3P72Q5  | KS57P21408<br>KS57P22416 | 100QFP              | SA-100QFP                                                        | Standard type |
| S3P72F5<br>S3P72P9  | KS57P21516<br>KS57P22532 | 100QFP              | SA-100QFP                                                        | Standard type |
| S3P7324             | KS57P3204                | 64QFP               | SA-64QFP                                                         | Standard type |
| S3P7335             | KS57P3316                | 80QFP               | SA-80QFP                                                         | Standard type |
| S3P7414/<br>S3P7434 | KS57P4104/<br>KS57P4304  | 42SDIP<br>44QFP     | SA-42SDIP<br>SA-44QFP                                            | Standard type |
| S3P7424             | KS57P4204                | 30SDIP<br>28SOP     | SA-30SDIP<br>SA-28SOP                                            | Standard type |
| S3P7515             | KS57P5116                | 64SDIP<br>64QFP     | SA57P5116-64SDIP(SA7515-64SDIP)<br>SA57P5116-64QFP(SA7515-64QFP) |               |
| S3P7528             | KS57P5208                | 42SDIP<br>44QFP     | SA-42SDIP<br>SA-44QFP                                            | Standard type |
| S3P7538/<br>S3P7533 | KS57P5308/<br>KS57P5312  | 30SDIP<br>32SOP     | SA-30SDIP<br>SA-32SOP                                            | Standard type |
| S3P7544             | KS57P5404                | 24SDIP<br>24SOP     | SA57P5404-24SDIP(SA7544-24SDIP)<br>SA57P5404-24SOP(SA7544-24SOP) |               |
| S3P7559             | KS57P5532                | 64SDIP<br>64QFP     | SA57P5116-64SDIP(SA7515-64SDIP)<br>SA57P5116-64QFP(SA7515-64QFP) |               |
| S3P7565             | KS57P5616                | 100QFP              | SA-100QFP                                                        | Standard type |
| S3P7574             | KS57P5704                | 64QFP               | SA-64QFP                                                         | Standard type |

## &lt; Appendix B &gt;

**SAM8(KS88P/Fxxxx, S3P/F8xxx Series) adapter socket table**

| <b>Device name</b>  |                           | <b>Package type</b>       | <b>Adapter socket (New name)</b>                                                                  | <b>Remark</b>      |
|---------------------|---------------------------|---------------------------|---------------------------------------------------------------------------------------------------|--------------------|
| <b>New</b>          | <b>Old</b>                |                           |                                                                                                   |                    |
| S3P80A5/<br>S3P80B5 | KS88P01016/<br>KS88P01116 | 24SDIP<br>24SOP           | SA-24SDIP<br>SA-24SOP                                                                             | Standard type      |
| S3P80E7             | KS88P01424                | 32SDIP<br>32SOP<br>40DIP  | SA88P0916-32SDIP(SA8095-32SDIP)<br>SA88P0916-32SOP(SA8095-32SOP)<br>SA88P0424-40DIP(SA8047-40DIP) |                    |
| S3P80F9/<br>S3P80G9 | KS88P01532/<br>KS88P01632 | 42SDIP<br>44QFP           | SA-42SDIP<br>SA-44QFP                                                                             | Standard type      |
| S3P8045/<br>S3P8047 | KS88P0416/<br>KS88P0424   | 32SDIP<br>32SOP<br>40DIP  | SA88P0916-32SDIP(SA8095-32SDIP)<br>SA88P0916-32SOP(SA8095-32SOP)<br>SA88P0424-40DIP(SA8047-40DIP) |                    |
| S3P8075             | KS88P0716                 | 64SDIP<br>64QFP           | SA-64SDIP<br>SA-64QFP                                                                             | Standard type      |
| S3P8095             | KS88P0916                 | 32SDIP<br>32SOP           | SA88P0916-32SDIP(SA8095-32SDIP)<br>SA88P0916-32SOP(SA8095-32SOP)                                  |                    |
| S3P821A             | KS88P2148                 | 80QFP<br>80TQFP           | SA-80QFP<br>SA88P2148TQFP(SA821A-80TQFP)                                                          |                    |
| S3P822B             | KS88P2264                 | 160QFP                    | SA-160QFP                                                                                         | Standard type      |
| S3F8235             | KS88F2316                 | 64SDIP<br>64QFP<br>64LQFP | SA-64SDIP<br>SA-64QFP<br>TBD                                                                      | Standard type(MTP) |
| S3P8245             | KS88P2416                 | 80QFP<br>80TQFP           | SA-80QFP<br>SA-80TQFP                                                                             | Standard type      |
| S3P8249             | KS88P2432                 | 80QFP<br>80TQFP           | SA-80QFP<br>SA-80TQFP                                                                             | Standard type      |
| S3P8454             | KS88P4504                 | 80QFP<br>80TQFP           | SA-80QFP<br>SA-80TQFP                                                                             | Standard type      |
| S3P8469             | KS88P4632                 | 64SDIP<br>64QFP           | SA-64SDIP<br>SA-64QFP                                                                             | Standard type      |
| S3P8475             | KS88P4716                 | 42SDIP<br>44QFP           | SA-42SDIP<br>SA-44QFP                                                                             | Standard type      |
| S3P84A4             | KS88P41004                | 64QFP                     | SA88P41004-64QFP(SA84A4-64QFP)                                                                    |                    |
| S3P8615             | KS88P6116                 | 42SDIP<br>44QFP           | SA88P6116-42SDIP(SA8615-42SDIP)<br>SA88P6116-44QFP(SA8615-44QFP)                                  |                    |
| S3P8629             | KS88P6232                 | 42SDIP<br>44QFP           | SA-42SDIP<br>SA88P6232-44QFP(SA8629-44QFP)                                                        |                    |
| S3P863A             | KS88P6348                 | 42SDIP<br>44QFP           | SA-42SDIP<br>SA88P6232-44QFP(SA8629-44QFP)                                                        |                    |
| S3F8647             | KS88F6424                 | 32SDIP                    | GA-32SDIP                                                                                         |                    |
| S3P8837             | KS88P8324                 | 42SDIP                    | SA88P8324-42SDIP(SA8837-42SDIP)                                                                   |                    |
| S3P8849             | KS88P8432                 | 42SDIP                    | SA88P8324-42SDIP(SA8837-42SDIP)                                                                   |                    |
| S3F880A             | KS88F80048                | 42SDIP                    | SA88P8324-42SDIP(SA8837-42SDIP)                                                                   |                    |

## &lt; Appendix C &gt;

**SAM8(KS86P/Fxxxx, S3P/F9xxx Series) adapter socket table**

| Device name         |                         | Package type                           | Adapter socket (New name)                                                                                                              | Remark                  |
|---------------------|-------------------------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| New                 | Old                     |                                        |                                                                                                                                        |                         |
| S3P9004/<br>S3P9014 | KS86P0004/<br>KS86P0104 | 40DIP                                  | SA86P0004-40DIP(SA9004-40DIP)                                                                                                          |                         |
| S3P921F             | KS86P20125              | 100QFP                                 | SA-100QFP                                                                                                                              | Standard type           |
| S3P9404             | KS86P4004               | 30SDIP<br>32SOP                        | SA-30SDIP<br>SA-32SOP                                                                                                                  | Standard type           |
| S3P9414             | KS86P4104               | 24SDIP<br>24SOP                        | SA-24SDIP<br>SA-24SOP                                                                                                                  | Standard type           |
| S3P9428             | KS86P4208               | 30SDIP<br>28SOP<br>32SOP               | SA-30SDIP<br>SA-28SOP<br>SA-32SOP                                                                                                      | Standard type           |
| S3P9434             | KS86P4304               | 16/18/20DIP<br>20SOP                   | SA-20DIP(16/18/20DIP)<br>SA-20SOP                                                                                                      | Standard type           |
| S3F9444             | KS86F4404               | 8DIP/8SOP                              | SA-20DIP/SA-8SOP                                                                                                                       | Standard type,<br>(MTP) |
| S3F9454             | KS86F4504               | 16/20DIP,<br>20SOP                     | SA-20DIP(16/20DIP)<br>SA-20SOP                                                                                                         | Standard type,<br>(MTP) |
| S3P9608             | KS86P6008               | 42SDIP<br>44QFP                        | SA-42SDIP<br>SA-44QFP                                                                                                                  | Standard type           |
| S3P9614             | KS86P6104               | 20DIP<br>24SOP                         | SA-20DIP<br>SA86P6104-24SOP (SA9614-24SOP)                                                                                             |                         |
| S3F9624*            | KS86F60204*             | 32SDIP                                 | TBD                                                                                                                                    | Standard type           |
| S3P9638             | KS86P6308               | 64SDIP<br>64QFP                        | SA-64SDIP<br>SA86P6308-64QFP (SA9638-64QFP)                                                                                            |                         |
| S3P9648             | KS86P6408               | 42SDIP<br>44QFP                        | SA-42SDIP<br>SA-44QFP                                                                                                                  | Standard type           |
| S3P9658             | KS86P6508               | 16DIP<br>18/20DIP<br>16SOP<br>18/20SOP | SA86P6508-16DIP(SA9658-16DIP)<br>SA86P6508-20DIP(SA9658-18/20DIP)<br>SA86P6508-16SOP(SA9658-16SOP)<br>SA86P6508-20SOP(SA9658-18/20SOP) |                         |
| S3P9664*            | KS86P6604*              | 20DIP<br>24SDIP<br>20SOP<br>24SOP      | SA9664-20DIP<br>SA9664-24SDIP<br>SA9664-20SOP<br>SA9664-24SOP                                                                          |                         |

\*Under-development device information is subject to change without notice

\*TBD: To be defined later.

## &lt; Appendix D &gt;

Others (S3P1860, S3FBxxx, S3F3xxx, S3F4xxx Series) adapter socket table

| <b>Device name</b> |              | <b>Package type</b> | <b>Adapter socket (New name)</b>        | <b>Remark</b>        |
|--------------------|--------------|---------------------|-----------------------------------------|----------------------|
| <b>New</b>         | <b>Old</b>   |                     |                                         |                      |
| Pellet (Chip)      | Pellet(Chip) | -                   | SA-Pellet                               | For on-board writing |
| S3P1860            | KS51P860     | 20DIP<br>20SOP      | SA1860-20DIP<br>SA1860-20SOP            | Special type (OTP)   |
| S3FB018            | KS85F00108   | 30SDIP<br>32SOP     | SA-30SDIP<br>SA-32SOP                   | Standard type (MTP)  |
| S3FB205            | KS85F2016    | 64QFP               | SA-64QFP                                | Standard type (MTP)  |
| S3FB41D            | KS85F40113   | 100QFP<br>100TQFP   | SA-S3FB41D-100QFP<br>SA-S3FB41D-100TQFP | Special type (MTP)   |
| S3FB42F*           | KS85F40225*  | 100QFP<br>100TQFP   | SA-100QFP<br>SA-100TQFP                 | Standard type (MTP)  |
| S3FB489*           | KS85F40832*  | 64SDIP<br>64QFP     | SA-64SDIP<br>SA-64QFP                   | Standard type (MTP)  |
| S3FB519            | KS85F50132   | 100QFP              | SA-100QFP                               | Standard type (MTP)  |
| S3F340F*           | KS17F40025   | 128QFP              | SA-128QFP                               | Standard type (MTP)  |
| S3F441F            | KS32F40125   | 64LQFP              | SA-S3F441FX                             | Special type (MTP)   |
| S3P4909            | KS32P6632    | 100LQFP             | SA-S3P4909-100LQFP                      | Special type (OTP)   |
| S3F380D            | KS17F80013   | 42SDIP              | SA380D-42SDIP                           | Special type (MTP)   |
|                    |              |                     |                                         |                      |

\* Under-development device information is subject to change without notice

\* TBD: To be defined later.